

Updated Tree of Thought – Key Entities & Relationships

This tree visualises the relationships between entities pertinent to the SKIIN Switzerland website narrative, incorporating additional insights from research and stakeholder feedback.

- **Heart Health Context**
- **Heart Disease** – leading cause of death; underscores need for screening.
- **Arrhythmias & AFib**
 - **Silent AF Episodes** – up to 70 % may be asymptomatic ¹ .
 - **Stroke Risk** – AF contributes to 20–30 % of ischemic strokes ² .
 - **Extended Monitoring Benefits** – multi-day monitoring detects far more arrhythmias than 24-h tests ³ .
- **Risk Factors** – chest pain, palpitations, family history, asymptomatic individuals.
- **Traditional Holter Problems** – uncomfortable, short duration, inconvenient hospital visits.
- **SKIIN Solution**
- **Products & Naming** – 10-Day Heart Screening, SKIIN 3X Screening™; self-pay options (3, 5, 10 days).
- **Technology**
 - **Textile Computing & Knitted Sensors** – conductive yarns and patented dry electrodes integrated into fabric via knitting; technology developed by **Myant** with contributions from **Nanoleq**, not ETH Zurich ⁴ .
 - **Carbon/Graphene Electrodes** – washable, hypoallergenic, durable ⁵ .
 - **Sensors Monitor Three Vital Signs** – heart rhythm, blood pressure trends, sleep breathing patterns.
 - **Comfort & Wearability** – soft garment; no wires or gels.
 - **Machine Washable & Reusable.**
- **Process**
 - **Eligibility Check** → **Referral** (GP or telemedicine) → **Delivery** → **Wear** → **Live** → **Return** (repack kit, apply prepaid return label) → **Analysis** → **Results.**
 - **Return Correction** – use prepaid return **label** included in box, not an envelope.
- **Platforms & Workflow**
 - **MVCP (Medical Vital Cloud Portal)** – admin portal for clinicians to set up studies and review real-time ECG data and symptom-by-symptom strips.
 - **Holter Analysis Software (Medical Algorithmics)** – AI pre-processing and technician review; cardiologists generate final diagnostic report.
 - **Secure Data Flow** – device → app → cloud → MVCP/analysis platform; compliant with GDPR/ Swiss law, MDR Class IIa.

- **AI + Human Review** – algorithms flag events; cardiac technicians and cardiologists interpret and confirm.

• **Audiences & Value Propositions**

- **Patients** – convenience, comfort, quick results, insurance coverage, emotional reassurance.
- **General Practitioners** – easy onboarding, no hardware cost, preventive screening in practice, strengthened referral relationships.
- **Cardiologists** – hospital-grade data, high compliance, extended monitoring yields, integration with EMR, separation of MVCP vs analysis.
- **Telemedicine Providers** – continuous data, API integration, liability reduction, multi-vital signs.
- **Corporate Wellness** – healthier employees, cost savings, volume discounts, anonymised reporting.

• **Emotional & Preventive Themes**

- Longevity & Quality of Life – “add years to life, life to years”.
- Preventive Health Urgency – silent arrhythmias necessitate screening.
- Arrhythmia–Stroke Link – emphasise risk but offer hope through early detection.
- Emotional Storytelling – personal vignettes (Maria, Thomas, Marc) illustrate impact.
- Caregiver Messaging – doing it for loved ones.
- **Medical Advisors** – Prof. Frank Ruschitzka, Prof. Matthias Wilhelm, Dr. Michiel Winter, Dr. Mehdi Namdar – authoritative endorsements.

• **Gaps & Corrections**

- Replace claims about ETH Zurich with accurate attribution to Myant/Nanoleq for sensor development ⁴.
- Correct return instructions to emphasise prepaid return **label**, not envelope.
- Distinguish **MVCP** (admin & live review) from **Holter Analysis software** (AI preprocessing and report generation).
- Avoid unsourced statistics (e.g., 95 % compliance); use qualitative descriptors or validated ranges.

This tree informs the updated specification, ensuring that new technical insights and corrections are woven into the messaging while preserving the overarching narrative of preventive, comfortable, and credible heart screening.

- 1 When Silence Isn't Golden: The Case of "Silent" Atrial Fibrillation - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC7252797/>

- 2 Atrial Fibrillation and Ischemic Stroke despite Oral Anticoagulation

<https://www.mdpi.com/2077-0383/12/18/5784>

- 3 Comparison of Arrhythmia Detection by 24-Hour Holter and 14-Day Continuous Electrocardiography Patch Monitoring - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC7220965/>

- 4 Myant acquires Swiss smart textile companies

<https://www.innovationintextiles.com/myant-acquires-swiss-smart-textile-companies/>

- 5 Carbon-Based Textile Sensors for Physiological-Signal Monitoring - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC10253857/>